

Customer Churn Prediction Using Machine Learning

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Abstract

Customer churn prediction has become an important area of study for organizations that seek to improve customer retention and reduce revenue loss. Churn occurs when customers discontinue the use of a company's products or services. This project presents a machine learning based customer churn prediction system developed using the IBM Telco Customer Churn dataset. Data preprocessing techniques such as handling missing values, feature transformation, and categorical encoding were performed before model training. Logistic Regression was used as the classification algorithm due to its effectiveness in binary classification problems. The developed model achieved an accuracy of approximately 78.75%. To improve accessibility and practical usability, a Streamlit based web application was deployed, allowing users to predict customer churn probability through an interactive interface.

Keywords: Customer Churn, Machine Learning, Logistic Regression, Streamlit, Classification, Predictive Analytics

I. Introduction

Customer retention is one of the most important challenges faced by businesses operating in competitive markets. Acquiring a new customer often costs significantly more than retaining an existing one. Therefore, predicting customer churn before it occurs enables organizations to take preventive measures and improve customer satisfaction.

Machine learning techniques have demonstrated strong capabilities in identifying hidden patterns within customer data. By analyzing historical customer information, predictive models can classify whether a customer is likely to continue using services or discontinue them.

This project develops a customer churn prediction system using machine learning techniques and deploys it as an interactive web application.

II. Problem Statement

Telecommunication companies often experience customer attrition due to competition, pricing issues, service quality concerns, and changing customer preferences. Manual identification of customers likely to churn is inefficient and prone to errors.

The objective of this project is to build an automated system capable of predicting customer churn using customer demographic and service-related information.

III. Objectives

The major objectives of this project are:

1. To analyze customer churn data.
 2. To preprocess and clean the dataset.
 3. To develop a machine learning classification model.
 4. To evaluate model performance.
 5. To deploy the model through a user-friendly web application.
 6. To provide churn probability predictions in real time.
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IV. Dataset Description

The IBM Telco Customer Churn dataset was used for this project.

Dataset Characteristics:

Parameter	Value
Total Records	7043
Total Features	21
Target Variable	Churn
Dataset Type	Structured Tabular Data

Important Features:

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Internet Service
- Contract Type
- Monthly Charges
- Total Charges

The target variable indicates whether a customer has churned from the service.

Figure 1. Dataset Preview

```
Columns: [c:\Users\SAUGATA BANERJEE\Downloads\archive\WA_Fn-UseC_-Telco-Customer-Churn.csv]
Index: []
PS C:\Users\SAUGATA BANERJEE\Desktop\ccproject> & "C:/Users/SAUGATA BANERJEE/AppData/Local/Python/pyt
honcore-3.14-64/python.exe" "c:/Users/SAUGATA BANERJEE/Desktop/ccproject/train_model.py"
(7043, 21)
```

	customerID	gender	SeniorCitizen	...	MonthlyCharges	TotalCharges	Churn
0	7590-VHVEG	Female	0	...	29.85	29.85	No
1	5575-GMVDE	Male	0	...	56.95	1889.5	No
2	3668-QPYBK	Male	0	...	53.85	108.15	Yes
3	7795-CFOCW	Male	0	...	42.30	1840.75	No
4	9237-HQITU	Female	0	...	70.70	151.65	Yes

```
[5 rows x 21 columns]
```

V. Methodology

The proposed methodology consists of five major stages.

A. Data Collection

The IBM Telco Customer Churn dataset was imported using the Pandas library.

B. Data Preprocessing

The following preprocessing steps were performed:

- Removal of unnecessary columns.
- Conversion of TotalCharges to numeric format.
- Handling of missing values.
- Encoding of categorical variables.
- Data cleaning and transformation.

C. Feature Engineering

Categorical variables were transformed into machine-readable numerical values using One-Hot Encoding.

The final processed dataset contained:

- 7043 records
- 31 transformed features

D. Model Training

The dataset was divided into training and testing sets using an 80:20 split.

The Logistic Regression algorithm was selected because:

- It performs well for binary classification.
- It is computationally efficient.
- It provides interpretable results.

E. Model Evaluation

The trained model was evaluated using prediction accuracy on the testing dataset.

VI. System Architecture

The workflow of the proposed system is shown below:

Dataset Collection

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Data Cleaning

↓

Feature Engineering

↓

Model Training

↓

Prediction Generation



Streamlit Web Application

VII. Tools and Technologies Used

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Scikit-learn
- Joblib
- Streamlit

Development Environment

- Visual Studio Code

Version Control

- Git
- GitHub

Deployment Platform

- Streamlit Community Cloud
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VIII. Results and Discussion

The Logistic Regression model achieved an accuracy of approximately:

78.75%

The developed prediction system successfully classifies customers as likely to churn or likely to stay.

Example Prediction:

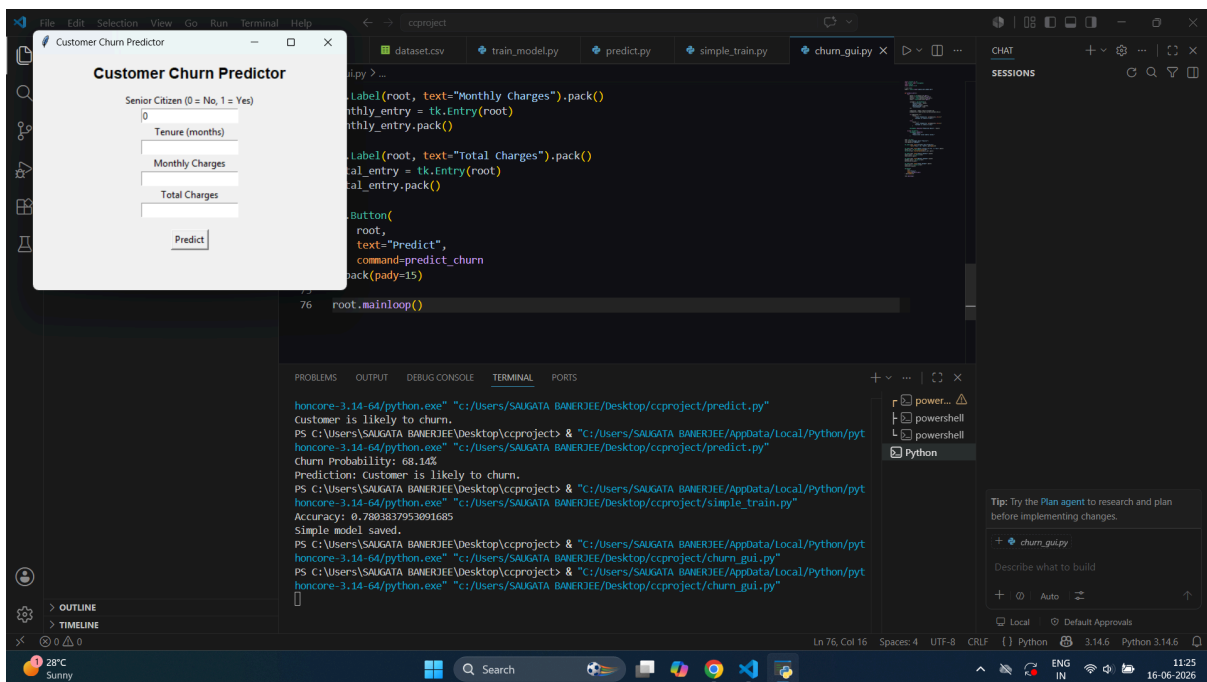
- Churn Probability: 68.14%
- Prediction: Customer is likely to churn

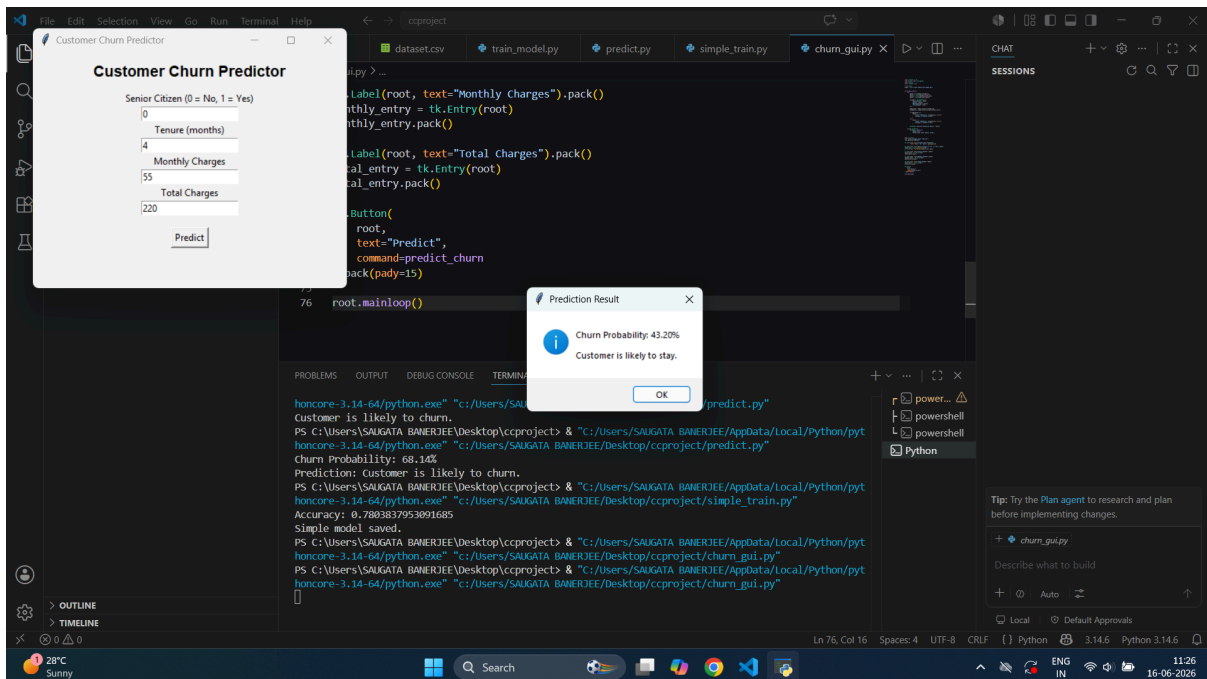
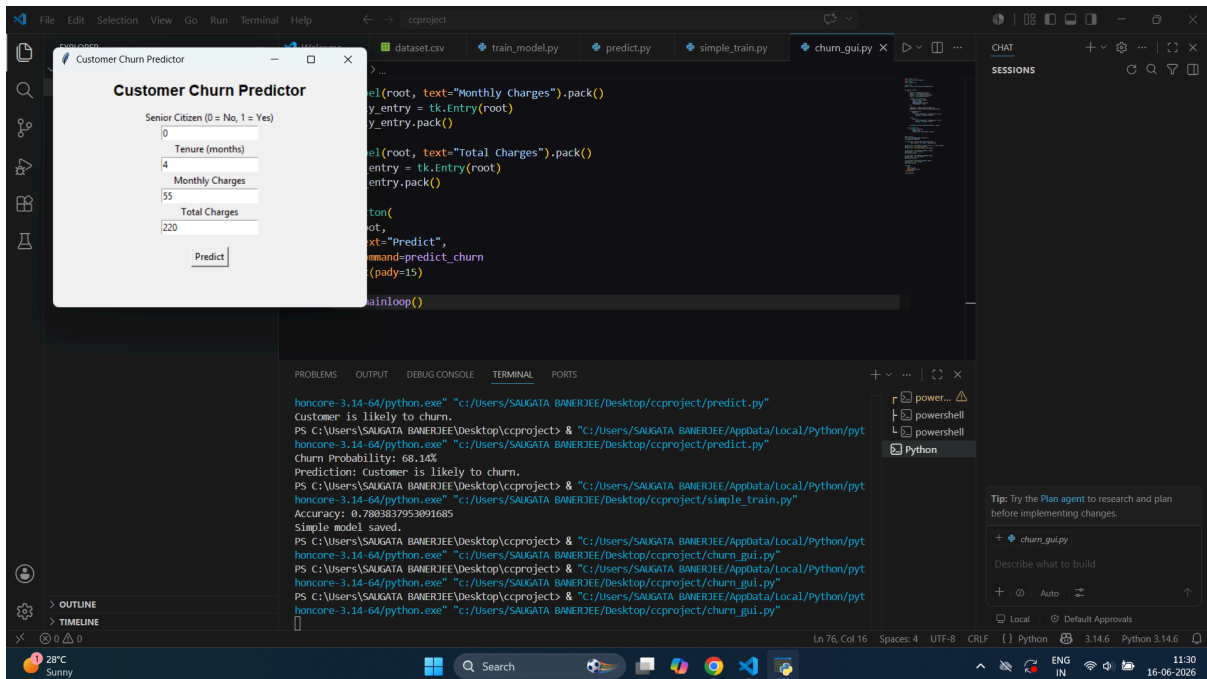
Figure 4. Model Accuracy

```
PS C:\Users\SAUGATA BANERJEE\Desktop\ccproject> & "C:/Users/SAUGATA BANERJEE/AppData/Local/Python/python-core-3.14-64/python.exe" "C:/Users/SAUGATA BANERJEE/Desktop/ccproject/train_model.py"
C:\Users\SAUGATA BANERJEE\AppData\Local\Python\python-core-3.14-64\Lib\site-packages\sklearn\linear_model\_logistic.py:599: ConvergenceWarning: lbfgs failed to converge after 1000 iteration(s) (status=1)
:
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=1000).
You might also want to scale the data as shown in:
  https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
  https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
Accuracy: 0.7874911158493249
Model saved successfully!
```

Figure 5. Sample Prediction Output





IX. Deployment

To improve accessibility, the machine learning model was deployed as a web application using Streamlit.

The application allows users to enter customer details and obtain churn predictions instantly.

Features:

- User-friendly interface
- Real-time predictions
- Churn probability estimation
- Public cloud deployment

Figure 6. Streamlit Application Interface

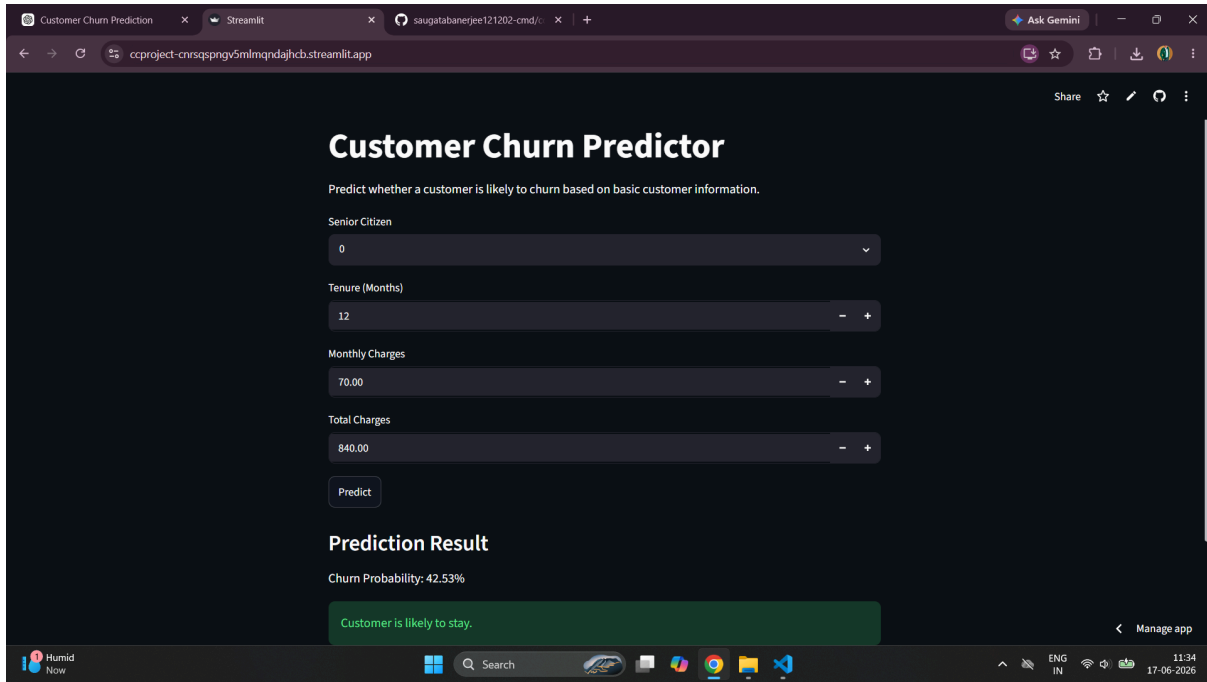
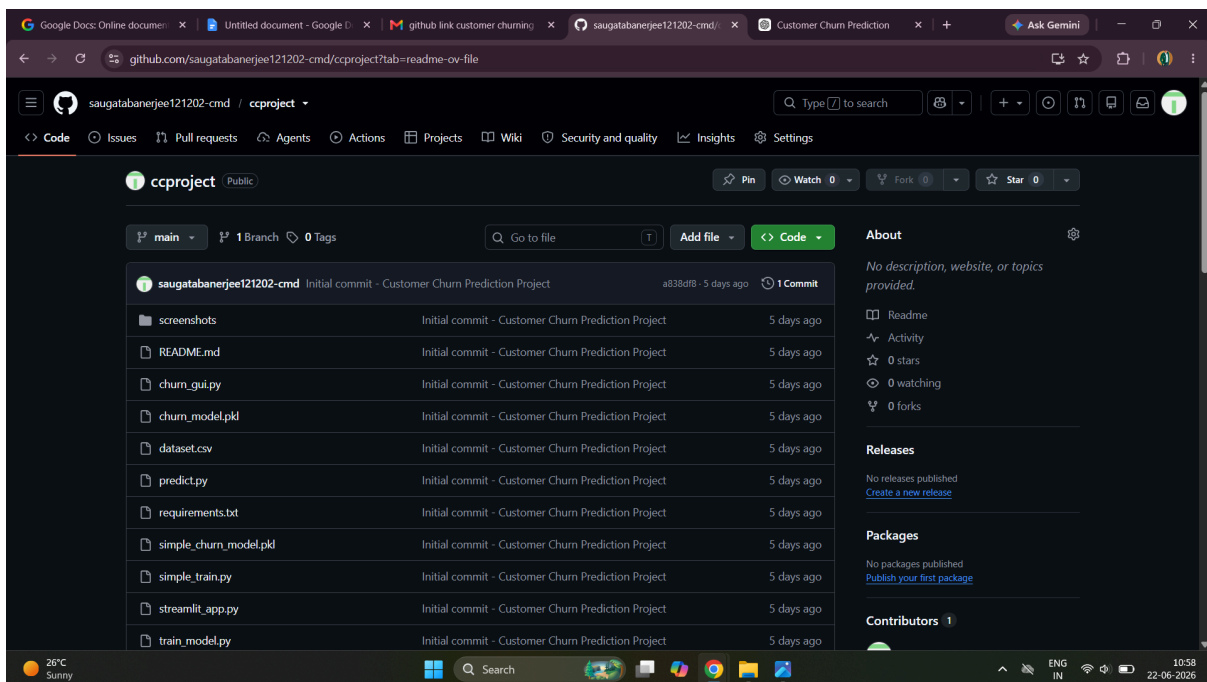


Figure 7. GitHub Repository



X. Conclusion

This project successfully developed a machine learning based customer churn prediction system using the IBM Telco Customer Churn dataset.

After performing data preprocessing, feature engineering, and model training using Logistic Regression, the model achieved an accuracy of approximately 78.75%.

The deployment of the model through Streamlit transformed the project from a standalone machine learning experiment into a practical web application. The system demonstrates the usefulness of machine learning techniques in solving real-world business problems and supporting customer retention efforts.

XI. Future Scope

Future enhancements may include:

- Implementation of Random Forest and XGBoost models.
 - Integration with real-time databases.
 - Addition of customer retention recommendations.
 - Visualization dashboards for business analytics.
 - Cloud-based enterprise deployment.
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References

[1] IBM Telco Customer Churn Dataset.

[2] Pedregosa et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research, 2011.

[3] Wes McKinney, Data Structures for Statistical Computing in Python, Python in Science Conference, 2010.

[4] Streamlit Documentation.

[5] Python Software Foundation Documentation.

[6] Scikit-learn Official Documentation.